Heterotopic Pancreas Masquerading as Omental Nodule on FAPI PET/CT

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Abstract

Chylous ascites, attributed to various etiologies including malignancy, tuberculosis, ruptured lymphatics, and congenital lymphatic disorders, manifests as abdominal distension. Our patient presented with this condition, and an elevated CA 125 prompted further investigation. Flourine-18 fluorodeoxyglucose positron emission tomography/computed tomography (PET-CT) revealed a metabolically inactive omental nodule, while gallium 68 fibroblast activation protein inhibitor (Ga-68-FAPI) PET-CT demonstrated uptake in the same nodule and low-grade uptake in bilateral adnexae. Colloid liver scan ruled out chronic liver disease. Surprisingly, lymphoscintigraphy showed no lymphatic leak. Histological examination of the omental nodule confirmed heterotopic pancreas (HP) in the small bowel mesentery, with normal adnexae. This case report illuminates the diagnostic challenges entailed in HP and signifies a pioneering instance in the literature where evidence of HP was identified for the first time on Ga-68-FAPI PET-CT during the investigative process.

Keywords: *CA 125, chylous ascites, flourine-18-fluorodeoxyglucose positron emission tomography/computed tomography, gallium 68 fibroblast activation protein inhibitor positron emission tomography/computed tomography, heterotopic pancreas*

Introduction

Heterotopic pancreas (HP) is characterized by anatomically separate tissue from main pancreas and has no vascular or ductal connections with it. It can be seen in stomach, small bowel mesentery, and rarely outside the gastrointestinal tract. Ectopic pancreas, pancreatic rest, accessory pancreas, aberrant pancreas, and pancreatic choristoma are synonyms. It is often incidentally diagnosed during endoscopy as submucosal swelling with central depression that represents the drainage of duct. Three types of HP cases have been described in the first histological classification system. The first (most common type) of HP tissue is composed of all the elements of the orthotopic pancreas, including acini, ducts, and islet cells. The second and third histologic types mainly consist of either acini or ducts. HP containing only islet cells can be described as the fourth histological type.

Fibroblast activation protein (FAP) is a type II transmembrane serine protease expressed in pathological conditions that include cancer, fibrosis, and inflammation. It expresses both dipeptidyl exopeptidase activity and endopeptidase activity that is not seen in normal fibroblasts.[1] Normal fibroblasts express dipeptidyl peptidase 4 expression with minimal or negligible FAP expression.

Gallium 68 FAP inhibitor (Ga68-FAPI) has high affinity for many tumors including colon, breast, thyroid, and lymphoma. It shows physiologic localization in uterus, urinary bladder, and sometimes pancreas.[2] Pancreas and salivary glands may show high dipeptidyl peptidase 4 expression and thereby localize Ga68 FAPI.[3,4]

We report a case of ectopic pancreatic tissue in the walls of jejunum that was masquerading as an omental nodule.

Case Report

A 41-year-old female presented with distension of abdomen. Her CA 125 was 164.7 u/mL and 265.4 u/mL at 3 months interval (normal <35). Computed tomography (CT) done previously revealed ascites, omental nodules, and thickening. Ascitic fluid was negative for malignant cells. However, it was rich in fat and triglycerides, i.e., chylous. A fluorodeoxyglucose positron emission tomography (FDG PET) CT

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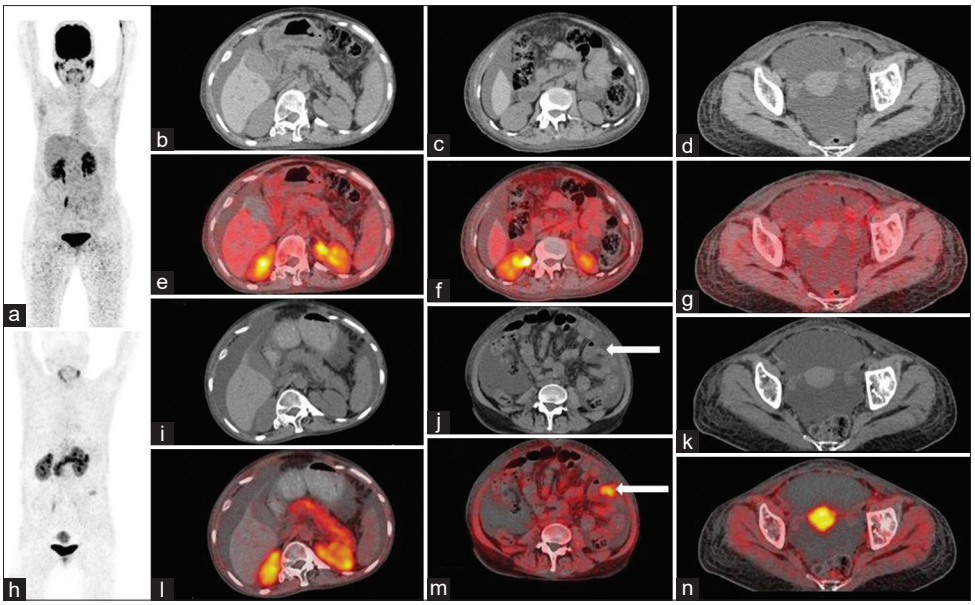


Figure 1: Fluorodeoxyglucose positron emission tomography computed tomography (FDG PET CT) shows no abnormal focal uptake in whole body MIP (maximum intensity projection) images (a), Pancreas shows no focus of abnormal FDG uptake (b and e), ascites is noted with no FDG avid omental or mesenteric nodules (c and f), bilateral adnexa are normal in size with no FDG uptake (d and g). 68Ga-Fibroblast activation protein inhibitor (68Ga-FAPI) scan shows physiologic uptake in kidneys, pancreas, bladder and uterus on MIP images (h), axial FAPI images confirm uptake in pancreas (i and l), FAPI avid omental nodule in left lumbar region (marked with white arrows in panels j and m), uterus with no abnormal FAPI uptake in adnexa (k and n)

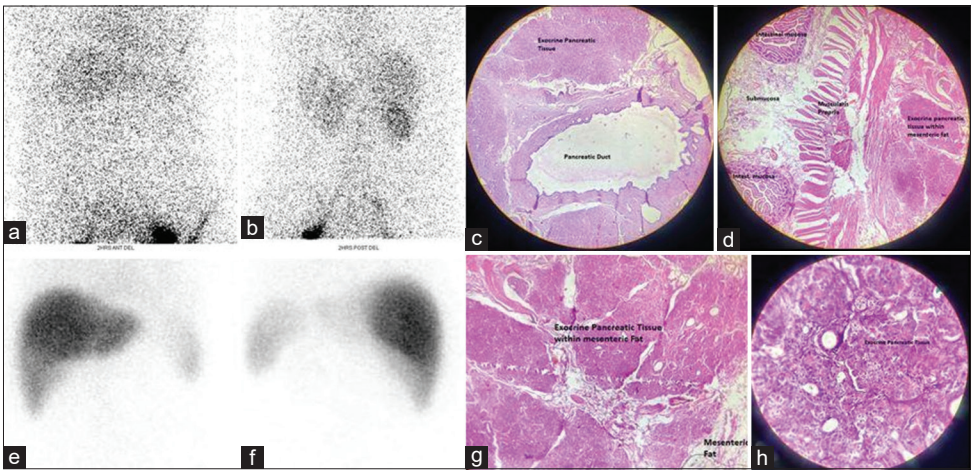


Figure 2: Lymphoscintigraphy does not show any abnormal leakage in abdomen (a and b), colloid liver scan shows physiologic distribution in the Kupffer cells of liver with no evidence of colloid shift, thereby ruling out chronic liver disease (e and f), excision of small bowel loop revealed ectopic pancreatic tissue (c, d, g, and h)

scan performed on Siemens Biograph 16-slice PET-CT system with a dose of 7.0 mCi did not show any focal hypermetabolic occult primary [Figure 1a-g]. In view of raised CA 125, Ga68-FAPI was performed on Siemens Biograph 16-slice PET-CT system with a dose of 3.0 mCi. There was a soft tissue lesion in omentum with FAPI uptake [Figure 1h, j, m]. Bilateral adnexae showed low-grade FDG and FAPI uptake [Figure 1k, n]. Pancreas shows physiologic uptake[Figure 1i, l]. Colloid liver scan was performed with 4mCi of 99mTechnetium phytate on Siemens Evo Excel Symbia Dual head Gamma camera using low energy high resolution collimator. It did not show any feature of cirrhosis [Figure 2e, f]. Lymphoscintigraphy performed with 0.5 mCi of 99m-Tc sulphur colloid on the same gamma camera did not show any evidence of lymphatic leak. Excision of the suspected omental nodule was performed with adjoining bowel loop [Figure 2a, b]. Histology of the nodule revealed ectopic pancreatic tissue in jejunal mesentery. Both the adnexa were free of malignancy [Figure 2c, d, g and h].

Discussion

Ascites can occur because of several causes that includes cirrhosis of liver, tubercular peritoneal involvement, congestive heart failure, nephrotic syndrome, pancreatitis, and ovarian-pancreatic malignancy.[5] Chylous ascites is accumulation of lipid-rich fluid in peritoneum. The

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triglyceride level in chylous ascites is more than 200 mg/dL.[6] The gold standard for chylous ascites is identification of chylomicrons using lipoprotein electrophoresis. However, this test is not readily available.

Chylous ascites can occur because of rupture of megalymphatics secondary to trauma or obstruction. Malignancy can cause fibrosis of a lymph node and leakage from dilated subserosal lymphatic into the peritoneum.[7] Malignancy and cirrhosis contribute two-third of chylous ascites.[8] Our patient was a middle-aged woman with elevated ovarian tumor marker and ascites. In view of suspected omental nodule, a diagnosis of adnexal primary was proposed. Ovarian neoplasms may be cold on FDG scan. Hence, a FAPI scan was performed which showed weak FAPI uptake in adnexa (SUV 1.6). However, there was a nodular FAPI uptake in the region of omentum as well (SUVmax 3.6).

Elevated levels of CA 125 have also been reported in chylous ascites.[9] Increased FDG uptake has been reported previously in heterotopic pancreatitis.[10] HP can cause complications such as inflammation, ulceration, bleeding, obstruction, intussusception, and rarely malignant transformation.[11] The imaging findings of HP are typical. Usually, the CT of the orthotopic pancreas creates a soft tissue-like appearance that mimics contours, lobulations, and enhancement similar to that of the mesentery. The contrast enhancement properties of HP may, in some cases, be different from orthotopic pancreas, which is due to the density of the contained acinus and islet cells. In these cases, it may be difficult to distinguish HP from lymphoma, gastrointestinal stromal tumor, leiomyoma, and accessory pathology.[12] At endoscopy as well, it can be mistaken for gastrointestinal stromal tumor, leiomyoma.

Normal pancreas does not show FDG uptake. FAPI may localize in variable degree in normal pancreas. FDG uptake in HP has been reported in stomach and esophagus.[13] However, despite extensive literature search, we did not find any report of FAPI uptake in HP. Histopathology remains the gold standard for the diagnosis.

The presence of heterotopic pancreatic tissue might trigger an inflammatory response, which can lead to increased vascular permeability and lymphatic vessel damage. The negative lymphoscintigraphy results, despite the presence of chylous ascites, may be elucidated through several conceivable reasons. First, the resolution limitations of lymphoscintigraphy may impede the detection of small or localized abnormalities in the lymphatic system. Second, given the intermittent nature of chylous ascites, negative results might arise if imaging coincides with periods of nonleakage. Third, alternative lymphatic drainage pathways may compensate for disrupted areas, evading detection by standard lymphoscintigraphy. Finally, technical factors during the imaging procedure could contribute to false negatives.

In summary, our case highlights the challenges in diagnosing chylous ascites with an elevated CA 125, especially when ovarian malignancy is suspected. Negative results on conventional imaging may necessitate the integration of novel modalities like FAPI PET-CT, showcasing its potential to provide critical insights in complex clinical scenarios.

Conclusion

This case report reveals the diagnostic challenges faced in a patient with a HP presenting as chylous ascites with raised CA 125, representing the first documented instance of the application of Ga-68-FAPI PET-CT in such a context. The identification of HP through Ga-68-FAPI PET-CT significantly advances our comprehension of this complex pathology.

Declaration of patient consent

The authors certify that they have obtained all appropriate patient consent forms. In the form, the patient(s) has/have given his/her/their consent for his/her/their images and other clinical information to be reported in the journal. The patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.

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Conflicts of interest

There are no conflicts of interest.

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